M.Tech Project Dissertation

on

Development of Digital Twin for Warehousing

Master of Technology

Stage - I Project Report

by

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Declaration

I affirm that this written submission is a representation of my own ideas, expressed in my own words. Wherever I have used the ideas or words of others, I have appropriately cited and referenced the original sources. I confirm that I have accurately acknowledged all sources used in the creation of this report. I also affirm that I have upheld all standards of academic integrity and honesty, ensuring that no idea, data, fact, or source has been misrepresented, fabricated, or falsified in this submission. I am aware that any breach of these principles may result in disciplinary action by the Institute and could also lead to legal consequences if sources are not properly cited or permissions were not obtained when required.

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Signature:

Date: October 22, 2026

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Dedicated to my Parents and my Late Grandfather

Abstract

Vision systems deployed in aerial, agricultural, and outdoor environments often degrade under adverse atmospheric conditions such as fog, smog, blizzards, or sandstorms. These effects cause significant distribution shifts in image statistics, reducing the reliability of vision transformers and multimodal encoders trained on clear-weather datasets.

This work explores diffusion-based generative frameworks to synthetically recreate such environmental conditions, enabling vision models to adapt and generalize effectively to real-world scenes.

A focused dataset was curated containing images of extreme atmospheric domains such as fog, smog, sandstorm, forest fire, storm, and underwater to evaluate existing, adaptive, and proposed solutions under consistent conditions.

Foundational diffusion architectures, including U–Net, DDPM and Conditional Diffusion were implemented to understand their generative dynamics. Building on these foundations, style-adaptive frameworks such as StyleShot, CSGO and InstantStyle were evaluated on the curated dataset using state-of-the-art vision transformers to assess their ability to reproduce physically consistent style shifts.

An existing model, **InstantStyle-Plus**, was adapted by extending InstantStyle with ResNoise and IP-Adapter modules for semantic and structural consistency, and further enhanced through a **Contrastive Style Descriptor (CSD)** guidance loop for perceptually aligned, distribution-shift–aware image synthesis .

Finally, a novel diffusion system integrating **InstantStyle with CSD loss** was introduced, guiding the denoising trajectory through contrastive style supervision to achieve physically realistic, depth-aware stylization while preserving semantic fidelity. The proposed framework demonstrates superior perceptual realism and environmental consistency, producing high-quality, style-shifted datasets that enhance the robustness of vision models for agricultural, aerial, and environmental monitoring applications.

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Chapter 1

Introduction

Images captured in high-altitude or extreme-weather regions such as fog, smog, dust storms, or snow often suffer from low contrast and color distortion. These degradations create a distribution shift between clean datasets and real-world imagery, causing transformer-based vision systems like ViT [7] and multimodal encoders such as CLIP [26] to perform unreliably. Most modern vision backbones, derived from the transformer paradigm [35], assume clear, well-lit scenes and therefore fail when deployed on aerial or drone cameras in low-visibility conditions. This project addresses that gap by generating synthetic data that emulate such adverse environments. Rather than relying on scarce real samples, diffusion-based generative models recreate realistic images exhibiting fog, cloud cover, pollution-induced smog, or heavy snow. By exposing vision networks to these synthetically shifted distributions, we aim to improve robustness and interpretability under natural domain variations, aligning with the broader generative learning frameworks that evolved from VAEs [16] and GANs [10]. Text and image-conditioned diffusion frameworks control the degree and type of degradation, enabling the creation of high-fidelity yet distribution-shifted datasets that support vision–language systems used in drone-based agricultural surveys at high-altitude.

1.1 Problem Statement

Current vision systems rarely encounter extreme environmental conditions and thus fail to generalize beyond clean training domains. This work focuses on *dataset synthesis* using diffusion models to generate realistic, distribution-shifted images under fog, mist, and snow. The goal is to design a controllable generation pipeline that yields diverse data and enables adaptation of pre-trained models to these scenarios, enhancing perception reliability in adverse weather. Collecting real high-altitude imagery is difficult and unsafe, while existing datasets remain scarce and unrepresentative. Generative modeling offers a practical way to replicate these conditions synthetically, producing realistic yet diverse images that preserve structure while incorporating degradations typical of extreme weather crucial for reliable performance in drone-based applications.

Chapter 2

Literature Survey

This chapter builds the theoretical foundation for the work presented in this report. It traces the evolution of modern generative modeling from attention-based architectures such as Transformers [35] and Vision Transformers [7] to early generative frameworks like GANs [10, 14] and VAEs [16], culminating in diffusion models that now dominate controllable image synthesis. The discussion then explores conditioning mechanisms, adapters, and perceptual style descriptors CLIP [26], IP-Adapter [3], and CSD [30] that make diffusion models both semantically and stylistically aware. The emphasis remains on clarity and intuition rather than exhaustive derivation.

2.1 Transformers: Attention-Based Sequence Modeling

Transformers [35] revolutionized neural network design by discarding recurrence in favor of parallel attention. Every element of a sequence can attend to every other, learning global context in a single step. The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V, \qquad Q = XW_Q,\ K = XW_K,\ V = XW_V.$$

Residual connections and layer normalization stabilize deep stacks [35], while positional encodings supply sequence order. Transformers became the blueprint for multimodal systems such as CLIP [26], Stable Diffusion [27], and IP-Adapter [3], where attention serves as the communication channel between modalities. Attention converts pairwise similarity into learnable weighting [35]. In diffusion models, this same principle governs how textual or visual prompts "speak" to the U–Net [28] at each denoising step.

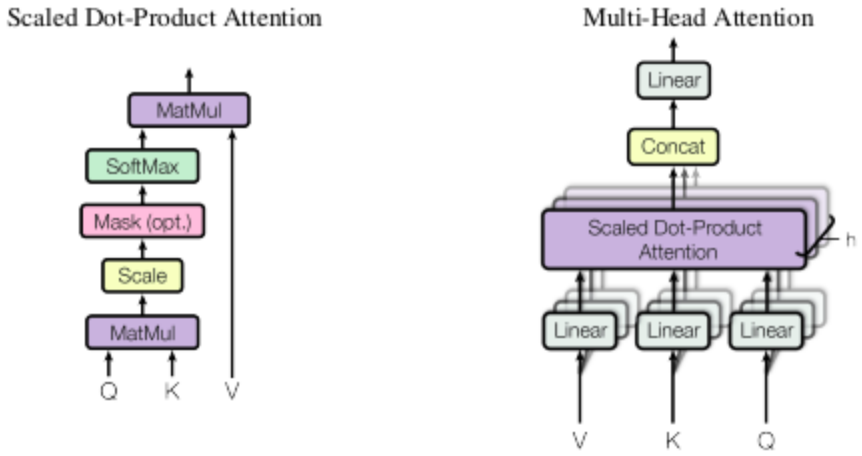


Figure 2.1: Self-attention pipeline: query–key similarity produces data-dependent mixing of values (adapted from Vaswani et al., 2017 [35]).

2.2 Vision Transformers (ViT)

The Vision Transformer (ViT) [7] treats an image as a sequence of non-overlapping $P \times P$ patches that act as tokens. Each patch is linearly projected and combined with positional embeddings before passing through stacked transformer encoders. A [CLS] token aggregates global information for classification.

ViT removes locality bias and allows any patch to attend to any other, thereby capturing long-range dependencies such as fog gradients or underwater scattering, which are critical for environmental style modeling.

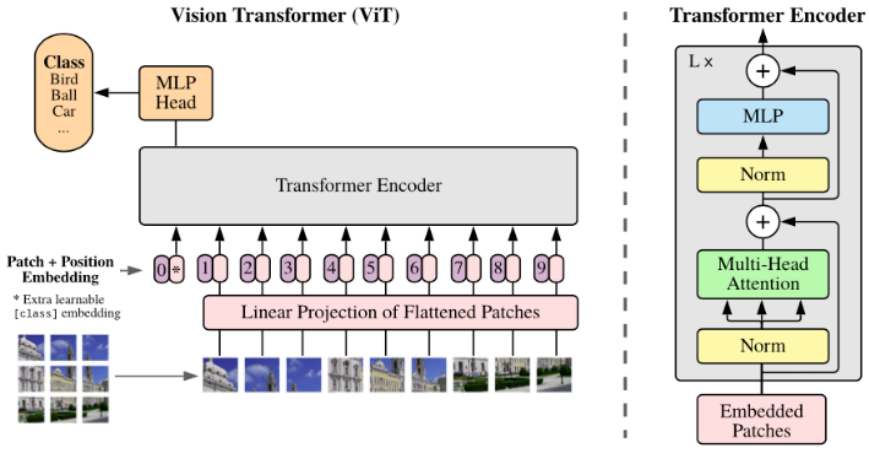


Figure 2.2: ViT pipeline and Transformer Block (adapted from Dosovitskiy et al., 2021 [7]).

2.3 Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) [10] were the first to produce high-fidelity synthetic images. They train a Generator G and Discriminator D in an adversarial loop:

$$\min_G \max_D \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))].$$

GANs introduced sampling from a learned distribution rather than memorizing examples, but adversarial training often causes instability and mode collapse. Variants such as WGAN [1] and StyleGAN [14] improved visual realism; however, they lacked semantic control and physical interpretability.

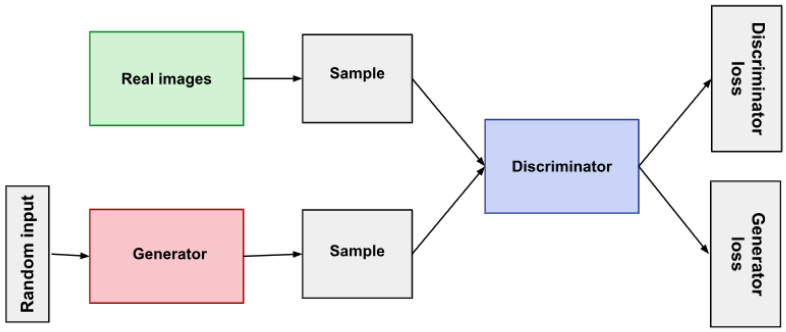


Figure 2.3: GAN architecture: generator–discriminator adversarial training loop (adapted from Foster, 2023 [8]).

2.4 Variational Autoencoders (VAEs)

VAEs [16] model data generation via a continuous latent variable z. They maximize the Evidence Lower Bound (ELBO):

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \text{KL}\big(q_\phi(z|x)\|p(z)\big), \quad z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon.$$

VAEs yield smooth, interpretable latent manifolds, enabling continuous interpolation between images. Although outputs can appear slightly blurred, VAEs provide stability and differentiable latent sampling, which are properties later exploited in Latent Diffusion Models (LDMs) [27].

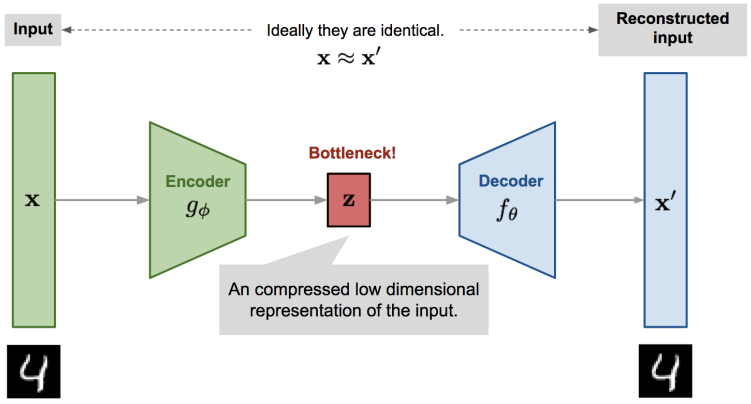


Figure 2.4: Variational Autoencoder: encoder–decoder with reparameterization trick (adapted from Foster, 2023 [8]).

2.5 U–Net Backbone for Vision and Diffusion

U-Net [28] provides hierarchical feature extraction through an encoder–decoder architecture with skip connections that preserve fine detail. Diffusion models augment U–Net with **time-step embeddings** (representing noise level) and **cross-attention layers** (for text or image conditions), forming the structural engine of modern denoising diffusion. The encoder progressively downsamples the noisy image into coarse latent features, capturing global semantics, while the decoder reconstructs spatial details through upsampling operations. Skip connections ensure that high-frequency information from the encoder is directly passed to the corresponding decoder stage, preventing the loss of geometric or boundary details. Time-step embeddings allow the model to adjust its denoising strength dynamically—focusing on coarse structure during early steps and fine details in later stages. Cross-attention integrates contextual guidance, enabling the model to align visual content with prompts, reference styles, or other conditioning inputs. This combination of hierarchical reconstruction, temporal awareness, and multimodal conditioning makes U–Net a robust backbone for diffusion-based image synthesis and environmental style transfer.

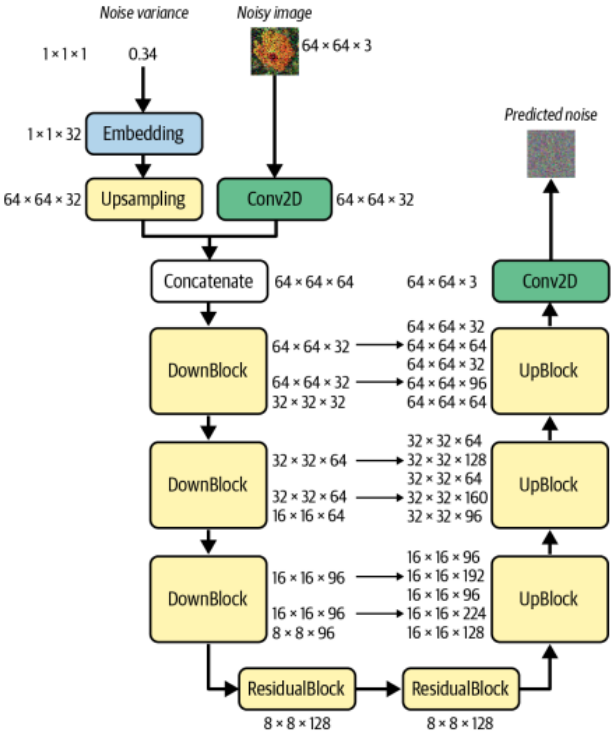


Figure 2.5: Diffusion U–Net: encoder–decoder with skip connections, time-step embeddings, and cross-attention (adapted from Foster, 2023 [8]).

2.6 Diffusion Models: From Noise to Structure

Diffusion models [12] constitute a family of generative models that gradually convert pure noise into structured data through a learned denoising process. They interpret image generation as the reversal of a known stochastic diffusion—a physically inspired process analogous to entropy reduction.

Forward (Noising) Process

The forward process progressively destroys image structure by adding Gaussian noise over T discrete timesteps according to a variance schedule $\{\beta_t\}_{t=1}^{T}$. Starting from a clean image x_0, each step produces a noisier version:

$$q(x_t|x_{t-1}) = \mathcal{N}\big(\sqrt{1-\beta_t}\, x_{t-1},\ \beta_t I\big).$$

Iterating this process yields a closed form for any timestep t:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i),$$

where $\epsilon \sim \mathcal{N}(0, I)$ represents random Gaussian noise. As $t \to T$, $\bar{\alpha}_t \to 0$ and x_T approaches pure noise, meaning all information about x_0 is effectively lost.

This stage mimics a physical diffusion process, where structured data dissolves into randomness. The corruption is fully known and mathematically invertible, allowing the model to learn the reverse process.

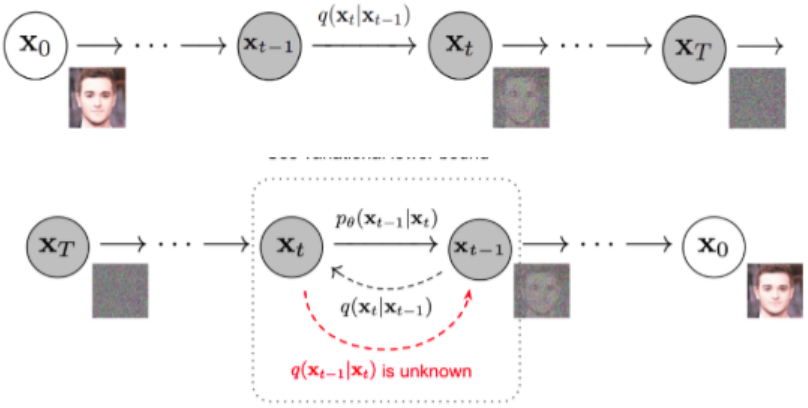


Figure 2.6: Forward and reverse diffusion: corruption by Gaussian noise and stepwise denoising (adapted from Foster, 2023 [8]).

Reverse (Denoising) Process

The reverse process attempts to invert the corruption and reconstruct the data distribution. A neural network $\epsilon_\theta(x_t, t)$ is trained to predict the noise component at each timestep. Given x_t, the model estimates ϵ_θ and computes the mean of the reverse conditional:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\, \sigma_t^2 I\big), \quad \mu_\theta(x_t, t) = \frac{1}{\sqrt{1-\beta_t}}\Big(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon_\theta(x_t, t)\Big).$$

Training minimizes the denoising objective:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,t,\epsilon}\big[\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\|^2\big].$$

Sampling. At inference, the process begins from random noise $x_T \sim \mathcal{N}(0, I)$ and iteratively applies

$$x_{t-1} = \frac{1}{\sqrt{1-\beta_t}}\Big(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon_\theta(x_t, t)\Big) + \sigma_t z, \qquad z \sim \mathcal{N}(0, I),$$

until x_0 emerges. Each denoising step restores a small fraction of structure, producing interpretable evolution from pure noise to a coherent image.

Observations

Diffusion can be understood as *entropy minimization* or *energy descent*. The forward process increases entropy (information destruction), while the learned reverse process systematically decreases it, similar to annealing a chaotic physical system back into order. This stepwise refinement makes diffusion both stable and explainable compared to GANs or VAEs.

Key Properties.

- **Stability:** Training minimizes a single, convex denoising objective, avoiding adversarial instability.
- **Interpretability:** Intermediate latents correspond to partially denoised states, revealing the generative trajectory.
- **Controllability:** External conditions (e.g., text, depth, style) can guide the reverse process at every step.

These properties make diffusion the most physically grounded and perceptually faithful generative paradigm, forming the basis for architectures such as Stable Diffusion [27], ControlNet [40], and the proposed CSD-guided models in this work.

2.7 Conditional and Guided Diffusion

To steer generation, diffusion can incorporate conditioning variables.

Classifier-Based Guidance. In this implementation [6] an external classifier $p_\phi(y|x_t)$ trained on noisy images guides sampling by adjusting the predicted noise:

$$\tilde{\epsilon}_\theta(x_t, t|y) = \epsilon_\theta(x_t, t) - w\,\sigma_t \nabla_{x_t} \log p_\phi(y|x_t),$$

where w scales guidance strength. This approach ensures class consistency but adds complexity and gradient noise.

Cross-Attention Conditioning. Modern diffusion integrates conditioning directly inside the U–Net [28]:

$$\text{CrossAttn}(Q, K, V) = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V,$$

with (K, V) from attention based [35] conditioning encoders (text or style) and Q from U–Net activations. This mechanism allows natural language or reference images to guide generation internally, forming the foundation of controllable diffusion.

2.8 Latent Diffusion and Stable Diffusion

Latent Diffusion Models (LDMs) [27] perform diffusion in the latent space of a pre-trained VAE, drastically reducing computation. **Stable Diffusion** [25] extends this concept by injecting CLIP [26] text embeddings via cross-attention, achieving text-to-image synthesis with photorealistic fidelity.

Operating in latent space focuses on perceptually relevant features. The text encoder becomes a compass that steers denoising toward the prompt description (for instance, "foggy valley" or "snow-covered forest").

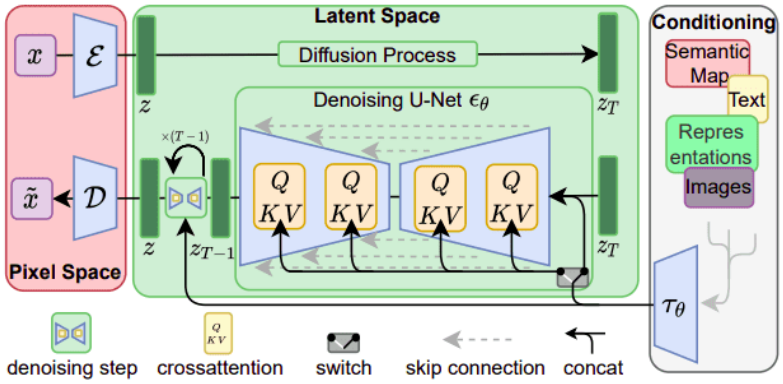


Figure 2.7: Stable Diffusion: text-conditioned latent denoising within a VAE encoder–decoder pipeline (adapted from Foster, 2023 [8]).

2.9 Contrastive Learning and CLIP Encoders

Contrastive learning aligns images and text in a shared embedding space. CLIP [26] employs dual encoders trained with the Information Noise-Contrastive Estimation loss:

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\log\frac{e^{v_i^\top t_i/\tau}}{\sum_j e^{v_i^\top t_j/\tau}} + \log\frac{e^{v_i^\top t_i/\tau}}{\sum_j e^{v_j^\top t_i/\tau}}\right].$$

Matching image–text pairs obtain high cosine similarity, giving diffusion linguistic grounding. However, CLIP conflates semantics and style, motivating newer decoupled adapters.

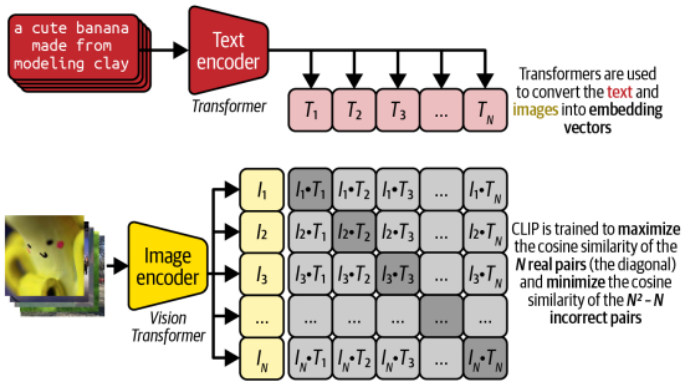


Figure 2.8: CLIP: contrastive alignment between image and text embeddings (adapted from Foster, 2023 [8]).

2.10 CSD: Contrastive Style Descriptors

Contrastive Style Descriptors (CSD) [30] define a perceptual embedding space purely for style. A ViT backbone learns to pull together stylistically similar pairs and push apart dissimilar ones. Similarity is computed as:

$$\mathrm{Sim}(x_g, x_r) = \frac{z_g^\top z_r}{\|z_g\|\|z_r\|}.$$

Unlike CLIP, CSD is content-invariant, capturing stylistic attributes such as haze intensity or lighting tone. In diffusion, a frozen CSD encoder can serve as a perceptual critic, aligning generated samples to the target style manifold.

If CLIP aligns "meaning," CSD aligns "feel." It measures how similar two images are in mood or visual atmosphere, thereby enabling perceptually guided diffusion.

2.11 Adapters, Structural Guidance, and Prompt Fusion

Adapters add modular control to pretrained diffusion models.

ControlNet [40] introduces a zero-initialized side network that processes structural cues (e.g., edges or depth) and merges them back into the U–Net to preserve geometry. It enables pixel-accurate layout control while maintaining stability. ControlNet acts as a geometry channel that learns to modulate generation based on structural hints such as sketch-to-image or depth-aware synthesis. The side network mirrors the layers of the main U–Net, ensuring that spatial information flows consistently across scales without disturbing the pretrained diffusion backbone. Because the parameters are initialized to zero, the model begins as an exact copy of the base diffusion system and gradually learns to incorporate external structure only when relevant. This design prevents training

instability and allows seamless integration of structural control without sacrificing prior semantic alignment. By conditioning on cues like canny edges, segmentation maps, or normal maps, ControlNet ensures that generated outputs adhere closely to desired spatial layouts while preserving natural texture and lighting. Its modular nature also allows multiple conditioning types—such as depth, pose, or scribbles—to be fused simultaneously, enabling precise and interpretable control over the generative process.

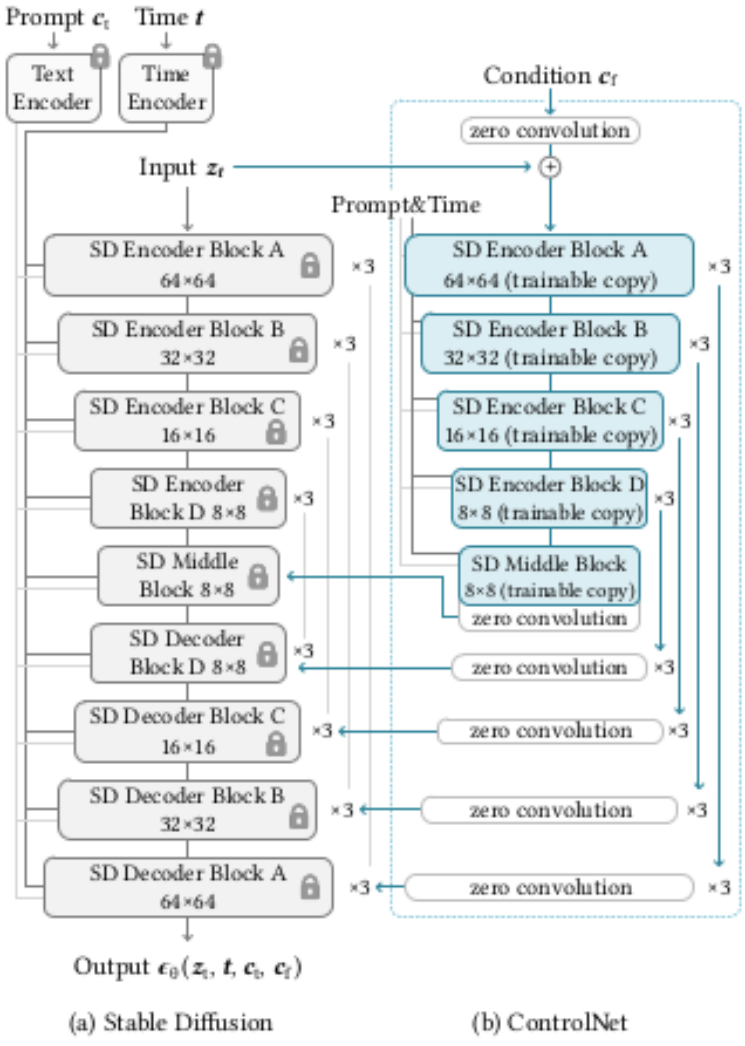


Figure 2.9: ControlNet: structure encoder merged into U–Net via additive connections (adapted from Zhang et al., 2023 [40]).

IP-Adapter [3] adds an image encoder whose embeddings are injected through decoupled cross-attention alongside text embeddings. InstantStyle [37] further refines this by subtracting the text-projected component of the image embedding, leaving a *style residual* for targeted appearance injection. This residual ensures that the visual

information contributes only stylistic cues—such as color tone, illumination, and material texture—while preventing semantic interference from the reference image. In the diffusion U–Net, the IP-Adapter's embeddings act as additional key–value pairs in cross-attention, influencing feature activations at selective layers where appearance dominates (typically early upsampling blocks). By decoupling semantics from appearance, the model can adapt the "look" of the generated scene while retaining the prompt's original structure and composition. This modular mechanism enables efficient and training-free personalization, allowing diffusion models to mimic a given visual domain (fog, underwater, dusk) simply by conditioning on a single reference image. Text dictates "what" to draw, whereas the reference image defines "how" it should appear—controlling tone, color, and texture without disturbing geometry or content alignment.

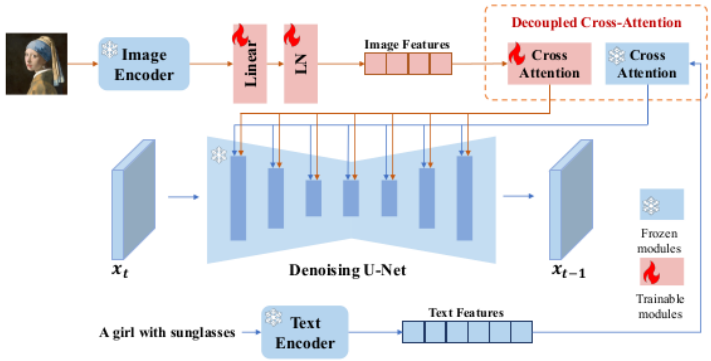


Figure 2.10: IP-Adapter: decoupled attention merges text semantics and image-based style cues (adapted from Chen et al., 2023 [3]).

Summary

Research has steadily advanced toward controllable and interpretable generation. Transformers introduced attention mechanisms, ViTs extended them to vision, and GANs and VAEs pioneered generative modeling. Diffusion unified these ideas into a stable probabilistic framework, enhanced by latent conditioning, cross-attention, and adapters like ControlNet and IP-Adapter. Contrastive systems such as CLIP and CSD added semantic and stylistic grounding, forming the basis for the adaptive, CSD-guided diffusion explored here.

Chapter 3

Foundational Experiments

This chapter presents a series of foundational experiments conducted to build an intuitive understanding of the key components that drive diffusion-based generative models. The intent was not to reproduce benchmark-level results but to practically observe how different architectures behave once trained, i.e., how the loss evolves, how reconstruction emerges from noise, and how various conditioning mechanisms influence the generation process. The experiments focused on the Vision Transformer (ViT) [7], the U–Net backbone [28], the Denoising Diffusion Probabilistic Model (DDPM) [12], and the Conditional Diffusion framework with classifier-guided generation [12, 26]. Together, they form the conceptual and empirical foundation upon which later adaptive and style-aware diffusion experiments are built.

Each experiment captures one essential property of diffusion systems: ViT illustrates the power of attention in capturing global context; U–Net demonstrates how spatial structure is preserved through skip connections; DDPM visualizes the transformation from noise to structure; and classifier-guided diffusion introduces controllability through explicit gradient-based conditioning.

3.1 Vision Transformer (ViT) Experiments

The Vision Transformer (ViT) [7] experiment was designed to understand how transformer-based vision models capture global relationships in images compared to convolutional architectures. A compact ViT with a shallow DETR-style [2] head was implemented to explore how patch embeddings interact with query-based decoding. Training was performed on CIFAR-10, with images resized to 224×224 to fit the patch embedding scheme. The setup used the Adam optimizer (3×10^{-4}), batch size 64, and 50 epochs of training with ReLU activation and 0.1 dropout. The network used a 768-dimensional embedding space, 12 attention heads, and a patch size of 16.

DETR (Detection Transformer) [2] was originally introduced to remove handcrafted detection stages such as anchor boxes or non-maximum suppression, relying solely on attention to perform end-to-end object detection. It formulates detection as a direct

set prediction task using the Hungarian loss for optimal bipartite matching between predicted and ground-truth objects. In this experiment, while detection itself was not the goal, integrating the DETR-style decoder provided a practical view of how attention can align semantically meaningful tokens, thereby learning structure without explicit convolution.

Results The ViT achieved around **85% test accuracy** on CIFAR-10. Attention visualizations clearly revealed that transformer heads concentrated on key object parts (e.g., airplane wings or vehicle bodies) while ignoring background clutter. Training was smooth, with stable convergence by the 40th epoch. The experiment reinforced the interpretability and global reasoning capacity of attention, thereby forming the mental framework used later to understand cross-attention in diffusion models.

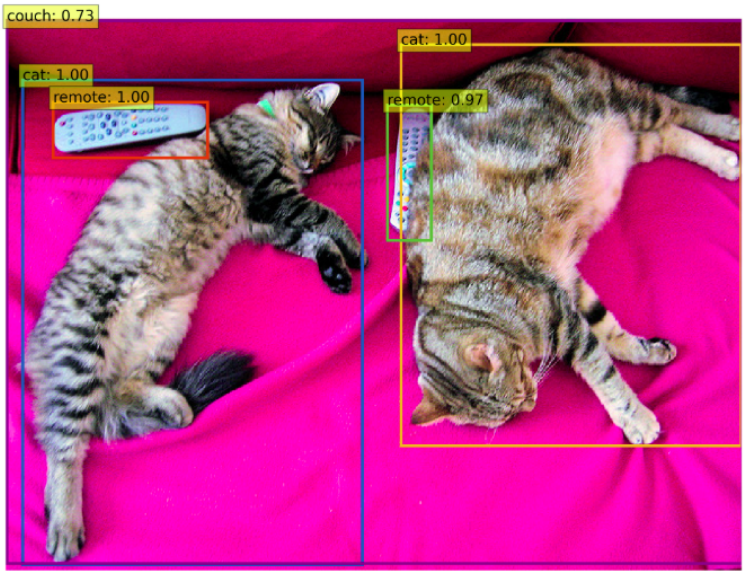


Figure 3.1: Vision Transformer on CIFAR-10: attention heads highlighting key semantic regions.

3.2 U–Net Backbone Experiments

Next, the U–Net [28] backbone was explored to understand how skip connections preserve fine structure during reconstruction, a feature that makes it the default denoising network in diffusion models. A basic U–Net autoencoder was trained on the Oxford-IIIT Pet dataset (128×128 resolution) for segmentation using Adam (1×10^{-3}), cross-entropy loss, and batch size 8. Each encoder block contained two 3×3 convolutions with ReLU followed by max pooling; the decoder mirrored this with transposed convolutions and symmetric skip connections. The architecture was intentionally kept simple to focus on reconstruction behavior rather than architectural sophistication.

Results Training converged quickly, with the model accurately recovering spatial boundaries after just a few epochs. Figure 3.2 shows clean, sharp segmentation outputs that closely match ground-truth masks. Skip connections clearly restored high-frequency information lost during downsampling, which explains why the same structure works so effectively in denoising networks. In diffusion settings, these skip pathways combine with sinusoidal time-step embeddings, thereby giving the model temporal awareness of which stage of denoising it is in and making U–Net both spatially and temporally adaptive.

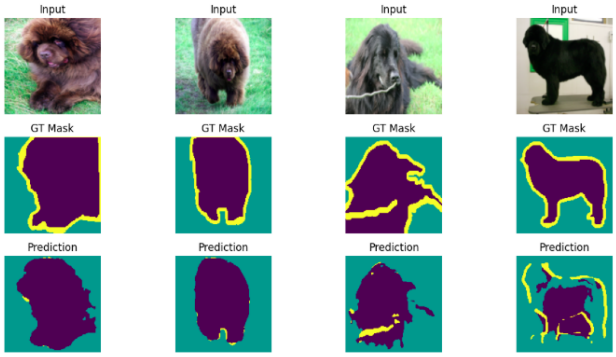


Figure 3.2: U–Net on Oxford-IIIT Pet: input, ground truth, and reconstructed output.

3.3 Denoising Diffusion Probabilistic Model (DDPM)

The DDPM [12] experiment visualized the *"order from noise"* process central to diffusion-based generation. A U–Net backbone was employed, augmented with sinusoidal time embeddings to encode temporal information. Training was performed on MNIST (resized to 32×32) for 15 epochs using the Adam optimizer (1×10^{-4}) and a linear β_t schedule ranging from 1×10^{-4} to 2×10^{-2} over 1000 diffusion steps. The network learned to predict the clean image x_0 from a noisy latent x_t by minimizing mean-squared error between the predicted and actual noise.

Results. Figure 3.3 illustrates the progressive refinement of generated digits. At the earliest diffusion steps, outputs resemble pure Gaussian noise with no discernible structure. As sampling proceeds, faint digit contours appear i.e a reflection of the model beginning to estimate global shape information. Midway through the denoising trajectory, the model stabilizes coarse structure (loops of 0, stalks of 1 or 9), while fine pixel-level details remain stochastic. In the final few steps, residual noise is systematically removed, revealing sharply formed digits with clear boundaries and correct topology.

This progression provides an intuitive visualization of how diffusion models learn to reverse entropy: rather than generating pixels directly, the model gradually transforms a noise distribution into structured data through a sequence of learned conditional transitions. The experiment also highlights the smooth and interpretable nature of diffusion learning unlike GANs [10], where generation can be abrupt or unstable, diffusion exhibits continuous improvement across time steps, offering clear insight into how the model internalizes spatial and probabilistic cues from the dataset. Despite being trained on a simple dataset (MNIST), the experiment effectively demonstrates the fundamental mechanics of diffusion, validating the theoretical framework that underpins more advanced models such as Stable Diffusion and InstantStyle.

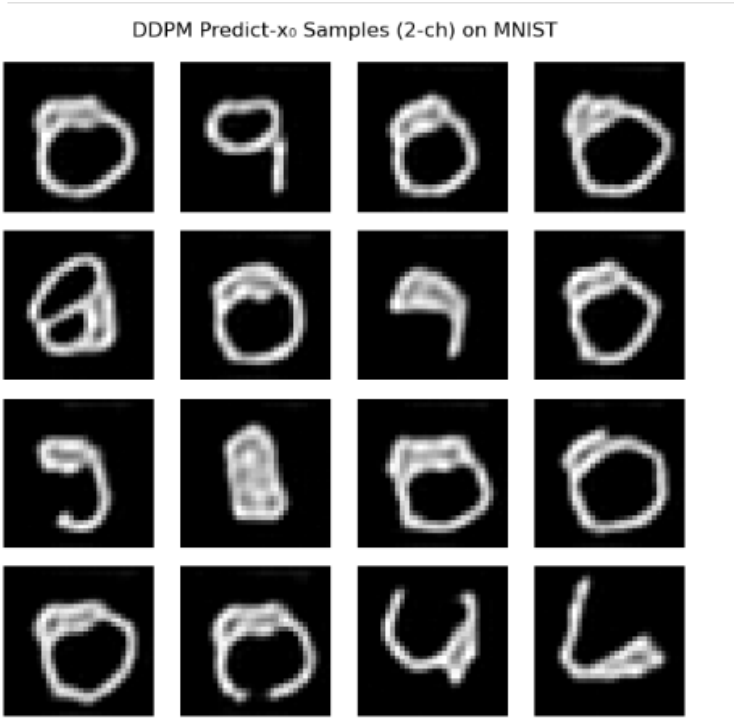


Figure 3.3: DDPM on MNIST: progressive denoising from pure noise to clear digits.

3.4 Conditional Diffusion (Classifier-Guided)

Finally, the classifier-guided diffusion setup introduced explicit control into the generative process. A DDPM denoiser was paired with a CNN [22] classifier trained on noisy MNIST samples to predict class labels at varying noise levels (Adam 1×10^{-3}, batch size 128, 10 epochs). During sampling, gradients from $\nabla_{x_t} \log p_\phi(y|x_t)$ were added to the DDPM mean estimate, scaled by a guidance factor $\lambda = 2.0$, thereby steering generation toward the specified target class.

Results At lower guidance scales, the model generated diverse samples, whereas higher λ values produced sharper and more class-consistent digits at the cost of variety. The results visually confirmed that classifier gradients can effectively modulate diffusion trajectories, which is a precursor to modern conditioning mechanisms such as cross-attention. This small but powerful experiment completed the conceptual bridge between unconditioned diffusion and controlled, prompt-driven generation.

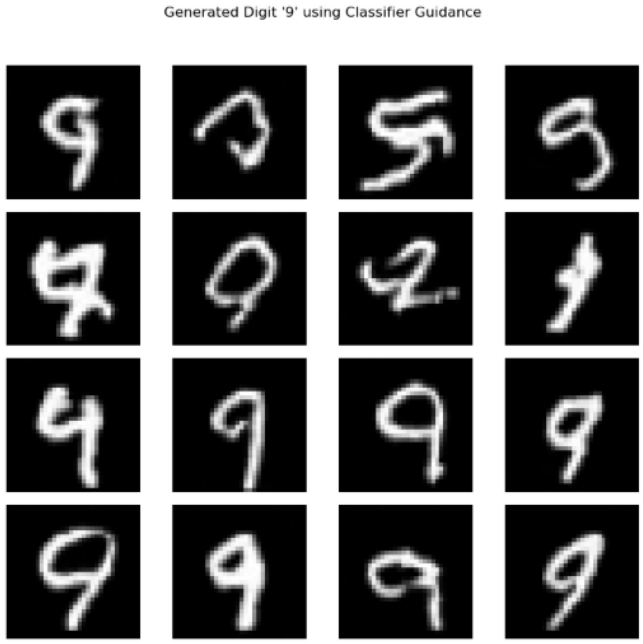


Figure 3.4: Classifier-Guided Diffusion on MNIST: class "9" generations at guidance scale $\lambda = 2.0$.

3.5 Summary of Foundational Experiments

The foundational experiments provided a cohesive understanding of the mechanisms behind diffusion-based image generation. The Vision Transformer [7] showed how attention captures long-range context, the U–Net [28] demonstrated spatial reconstruction, and the DDPM [12] revealed structure emerging from noise, while conditional diffusion established controllability through gradient-based guidance. Together, these results grounded the theory and laid the foundation for exploring style-aware and adaptive diffusion frameworks such as StyleShot [9], CSGO [39], InstantStyle [37], InstantStyle-Plus [36], and the proposed CSD [30]-guided architecture.

Chapter 4

Experiments on Existing Solutions

Before evaluating any existing style-aware diffusion systems, a focused dataset was curated to represent the kind of distribution shifts observed in the wild. Approximately **500 images** were handpicked from publicly available sources such as Pinterest [24], iStock [13], and Google Images [11], spanning a wide range of adverse environmental conditions such as fog, smog, heavy rain, snow, blizzard, sandstorm, dust storm, storm, and forest fire. These scenes were specifically selected because both the background and foreground elements are often veiled under a strong stylistic "shroud," which makes object boundaries ambiguous and significantly complicates classification and detection tasks. This dataset serves as the common evaluation ground for all subsequent models. **Note:** Although testing and benchmarking were carried out across all environmental domains mentioned above, the following visual comparisons and analyses focus specifically on the **agriculture domain**, which was found to most effectively resonate with the objectives of realistic distribution-shift representation and volumetric environmental stylization.

4.1 StyleShot: One-Shot Style Transfer with Attention Modulation

To explore prompt-guided, reference-based style transfer under environmental distribution shifts, the first existing solution evaluated was **StyleShot** [9]. The experiments were carried out on the curated dataset mentioned above. These scenes were selected to intentionally represent the loss of visibility and structural ambiguity typical of style-induced distribution shifts, where both the background and foreground appear enveloped in a dense "style shroud." The goal was to examine whether the model could convincingly reproduce such styles while maintaining semantic coherence.

Architecture Overview As illustrated in Figure 4.2, the original StyleShot framework is designed as a **tri-modal diffusion system**, integrating three modalities: text, content, and style into a unified diffusion backbone. It consists of three encoders: (1) a

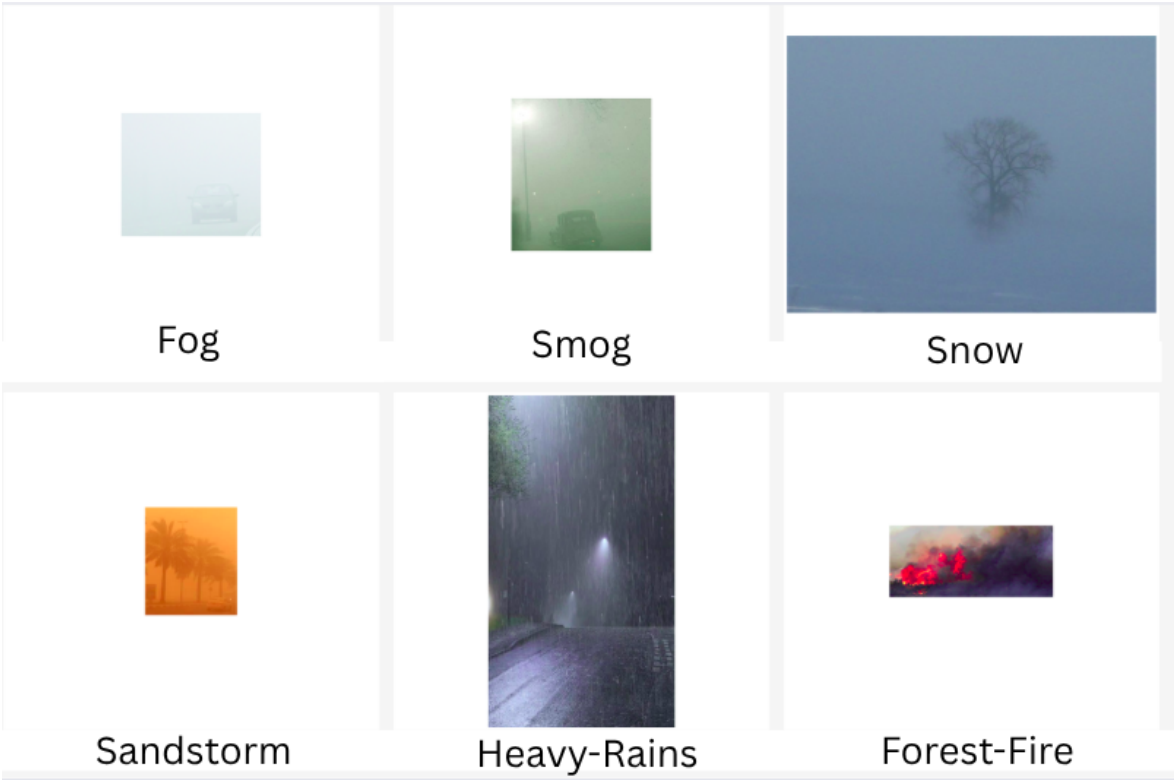


Figure 4.1: Representative collage from the curated dataset used for evaluation, spanning diverse environmental conditions such as fog, smog, snow, sandstorm, and forest fire. The agriculture domain subset (shown here) was selected for focused visual analysis due to its strong correspondence with real-world visibility and scattering effects.

Style-Aware Encoder that partitions the style reference image into multiple scales and embeds each partition using transformer blocks (OpenCLIP ViT-H/14) [26] to capture tone, color, and texture statistics; (2) a **Content-Fusion Encoder** that processes edge maps or structure references to maintain spatial layout and (3) a **CLIP Text Encoder** that provides semantic grounding for prompt-based control. These embeddings are fused within the diffusion U–Net [28] via **Parallel Cross-Attention** and **Residual Addition**, enabling the network to modulate appearance (from the style image) while preserving textual and structural semantics .

Modified Experimental Configuration In this work, the setup was simplified to reflect the **text + style reference** configuration, as the curated dataset did not include explicit content reference images or edge maps. Accordingly, the *Content-Fusion Encoder* branch was disabled. The model thus operated as a **bi-modal diffusion system**, driven by the CLIP text embeddings for semantic composition and the style-aware embeddings for texture and tone modulation. While the paper depicts the full tri-modal fusion (Figure 4.2), in our experimental setting only the *Style-Aware Encoder*

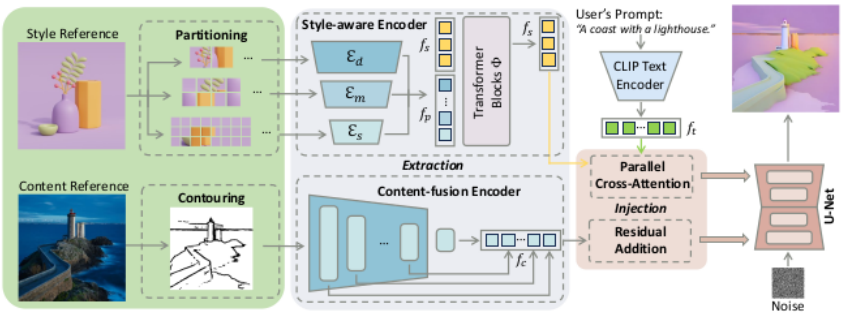


Figure 4.2: Overall architecture of StyleShot [9]: tri-modal fusion of style, content, and text embeddings through parallel cross-attention and residual addition in the diffusion U–Net.

and *CLIP Text Encoder* contributed to conditioning. The fusion occurred through parallel cross-attention layers within the diffusion U–Net, where style tokens (f_s) and text embeddings (f_t) jointly influenced the denoising trajectory:

$$\text{Attn}_{\text{final}} = \text{Attn}(Q, K_t, V_t) + \lambda_s \, \text{Attn}(Q, K_s, V_s),$$

with λ_s controlling the relative style strength. This configuration aligns with the objectives of this study, which involve text-driven style transfer under distribution shift, where the style reference image represents an environmental condition such as haze, snow, or dust.

Results When evaluated on the curated dataset, StyleShot [9] generated outputs that were visually rich but **overly artistic** and aesthetically biased rather than physically realistic. The model often prioritized stylistic exaggeration over faithful reconstruction, producing textures reminiscent of oil painting or digital art. This deficiency became evident in agricultural and outdoor scenes, where machinery and terrain edges remained sharply defined while the background appeared painterly and abstract. Similarly, in underwater or snow-heavy conditions, StyleShot was unable to simulate realistic scattering, depth-based attenuation, or backscatter, producing flattened tones that lacked the characteristic bluish-green absorption seen in real imagery.

Inference StyleShot offers an elegant and training-free mechanism for reference-based style transfer but demonstrates clear limitations when applied to environmental domain adaptation tasks. Its diffusion outputs are dominated by artistic stylization, fail to produce realistic haze or scattering around the foreground, and often degrade structural fidelity. These shortcomings motivate the exploration of more adaptive, physics-aware frameworks such as InstantStyle [37], CSGO [39], and the proposed CSD [30] guided approach in later sections.

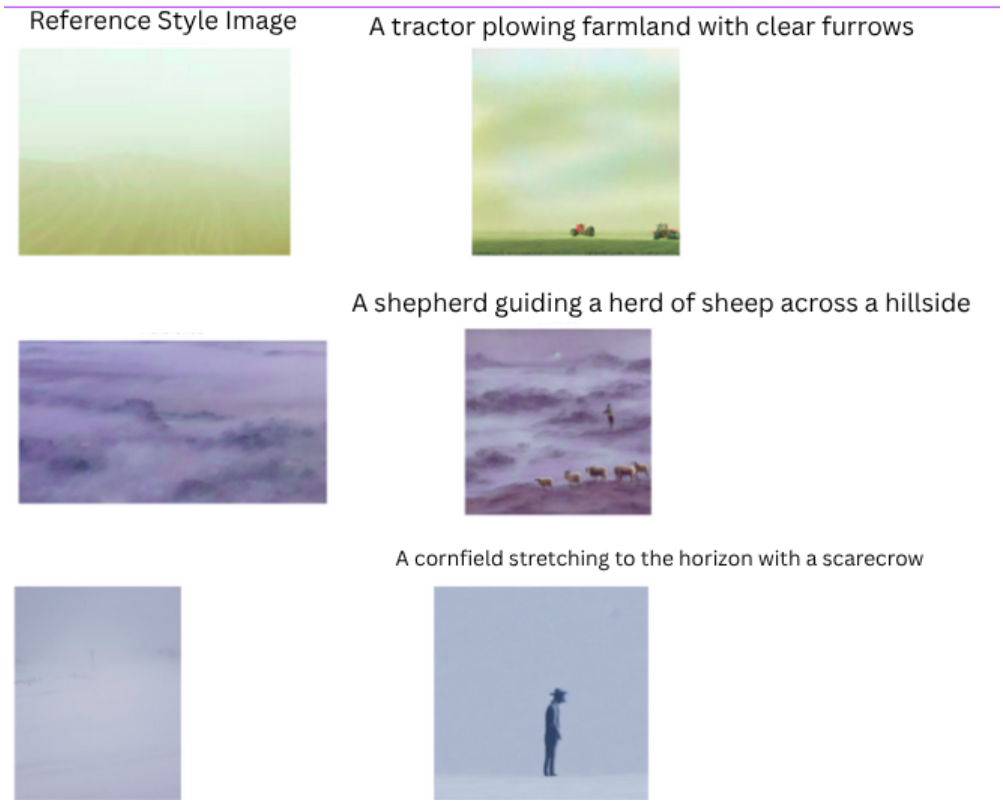


Figure 4.3: StyleShot on curated agricultural scenes: over-stylization and lack of depth-consistent haze. Foreground regions remain overly sharp while the background becomes painterly.

4.2 CSGO: Content–Style Composition in Text-to-Image Generation

To assess an end-to-end trained alternative to tuning-free stylization, we evaluated **CSGO (Content–Style Composition)** [39] on the curated extreme-conditions set. The objective remains to transfer the environmental "style shroud" while preserving scene semantics and producing physically plausible depth-dependent effects.

Architecture Overview and Configuration CSGO is built on an SDXL backbone [25] and explicitly *decouples* content and style controls. A *content pathway* injects CLIP image encoder features [26] via cross-attention in the down blocks and a *Tile ControlNet* stream [40] in the up blocks. In parallel, a *style pathway* encodes the reference style image using a ViT-H encoder [7] and a Perceiver Resampler, injecting tokens into both the U–Net and ControlNet. Training uses the **IMAGStyle** dataset (210k triplets) with content from MSRA10K/MSRA-B and styles from WikiArt and Midjourney.

In this study, CSGO was run in a **text + style** regime (no content reference). A

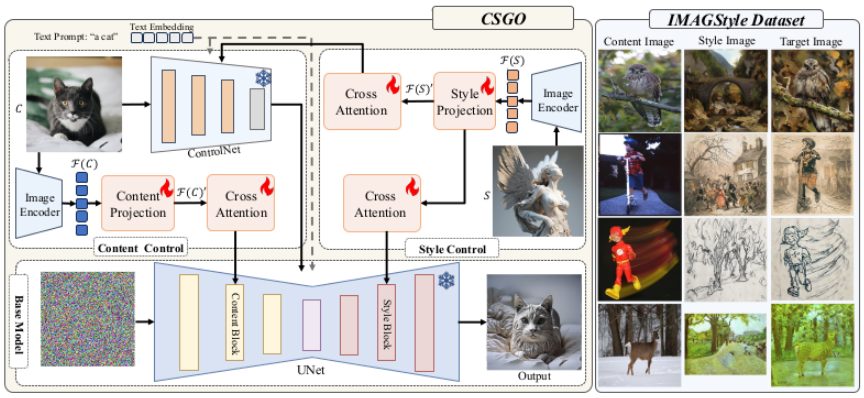


Figure 4.4: Overview of the CSGO framework: decoupled content and style pathways fused into SDXL with ControlNet support (adapted from Xing et al., 2024 [39]).

blank canvas was provided as content input; prompts defined semantics, and the style image supplied appearance tokens. ControlNet (Tile) was used at low conditioning scale (0.2). Sampling used 50 steps, guidance 5.5, `style_scale`=1.0, `content_scale`=1.3.

Results CSGO delivered strong stylistic color bias but lacked physics-consistent environmental haze. Foregrounds stayed sharp while backgrounds were softly recolored, failing to create volumetric occlusion. Underwater and fog scenes lacked depth-dependent attenuation and scattering, producing flatter results.

Inference CSGO provides a solid framework for content–style separation but, in a text+style setup, lacks physical realism. Without a content image prior or strong ControlNet, style behaves like recoloring rather than volumetric shrouding. This motivates physics-aware guidance in later models.

4.3 InstantStyle: Style-Preserving Injection with IP-Adapter

Among all evaluated methods, **InstantStyle** [37] consistently delivered physically grounded and semantically coherent text-guided style transfer across adverse environmental domains. As a tuning-free adapter-based framework built on SDXL [25], it introduced a minimalist yet powerful mechanism for injecting style residuals directly into the diffusion process without modifying the base network weights. This approach offered the best trade-off between *semantic consistency, structural fidelity*, and *visual realism*, setting the stage for adaptive variants such as InstantStyle-Plus [36].

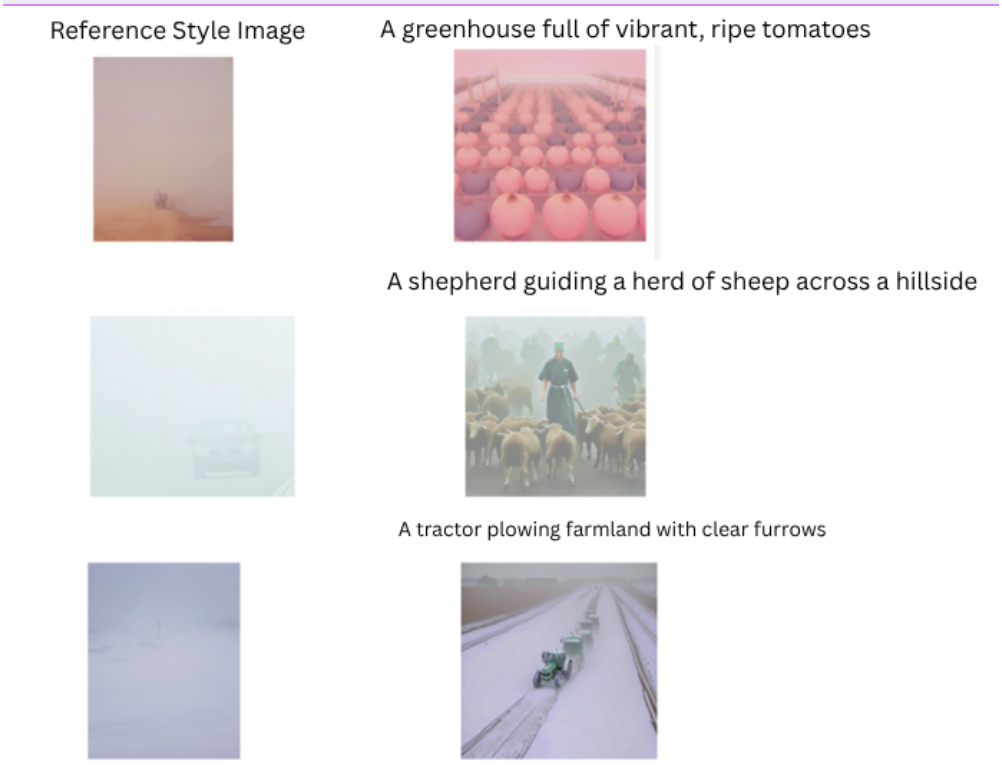


Figure 4.5: CSGO on curated agricultural scenes (text+style setup): texture transfer without depth-consistent haze; crisp foreground edges.

Architecture and Embedding Formulation InstantStyle augments the SDXL backbone with an additional *Image Prompt Adapter* (IP-Adapter [3]) that encodes the style image in the same latent CLIP [26] space as the text embedding. Given a text prompt t and a style image I_s, the system computes the following CLIP embeddings:

$$\mathbf{e}_t = f_{\text{text}}(t), \qquad \mathbf{e}_s = f_{\text{img}}(I_s),$$

where $f_{\text{text}}(\cdot)$ and $f_{\text{img}}(\cdot)$ denote the text and image encoders of the pre-trained CLIP model respectively. To disentangle appearance from semantics, InstantStyle applies a simple but effective subtraction in the CLIP feature space:

$$\mathbf{e}_{\text{style}} = \mathbf{e}_s - \beta\, \mathbf{e}_t,$$

where β is a scaling coefficient (typically $\beta \in [0.6, 0.8]$) controlling the degree of content suppression. This residual embedding $\mathbf{e}_{\text{style}}$ encodes pure stylistic cues—color palette, illumination, texture statistics—while discarding semantic overlap with the text description.

The resulting style tokens are linearly projected into the attention space of SDXL's U–Net [28] via the IP-Adapter, forming a set of key–value pairs:

$$(\mathbf{K}_{\text{style}}, \mathbf{V}_{\text{style}}) = \mathcal{P}(\mathbf{e}_{\text{style}}),$$

where $\mathcal{P}(\cdot)$ denotes the learned style projection module. These pairs are injected into selected mid- and up-block cross-attention layers, replacing or complementing the text-based conditioning. Thus, for an intermediate latent feature $\mathbf{z}_l$ at layer l, the attention update becomes:

$$\text{Attn}_l(\mathbf{z}_l) = \text{softmax}\left(\frac{\mathbf{Q}_l \left[\mathbf{K}_t \parallel \mathbf{K}_{\text{style}}\right]^\top}{\sqrt{d}}\right) \left[\mathbf{V}_t \parallel \mathbf{V}_{\text{style}}\right],$$

where $\parallel$ denotes concatenation along the token dimension, ensuring that both semantic and stylistic contexts influence the generative trajectory.

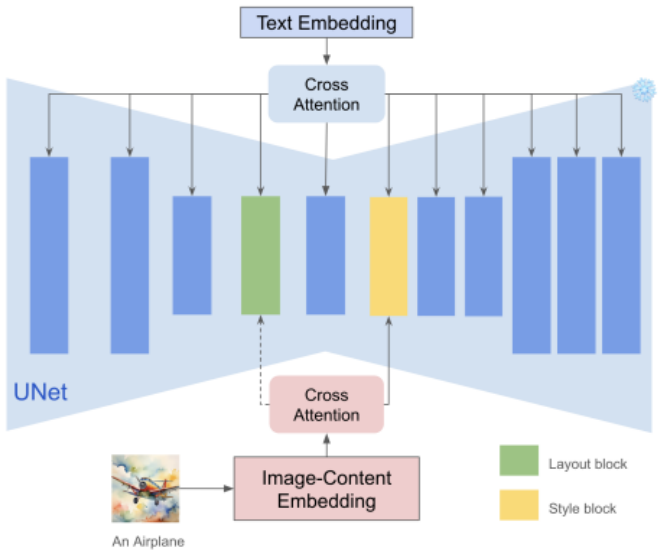


Figure 4.6: InstantStyle architecture: CLIP-based text and style embeddings, residual subtraction, and IP-Adapter cross-attention injection into SDXL layers (adapted from Wang et al., 2024 [37]).

Configuration and Modes We evaluated two principal modes:

- **Soft Mode:** 40 diffusion steps, guidance 6.8, and style scale 0.83 — prioritizing volumetric realism and semantic faithfulness.
- **Strong Mode:** Higher guidance and style scale, emphasizing contrast and saturation at the expense of depth consistency.

Additionally, a **tuned-soft** variant relaxed anti-blur terms in the negative prompt, such as "sharp focus" and "deformed," to allow physically consistent diffusion of light and haze in weathered environments.

Results Across the agriculture domain, InstantStyle's tuned-soft configuration produced perceptually rich and physically consistent results. The model achieved volumetric haze and subtle scattering effects while retaining the structural geometry of the base content. Foreground objects remained coherent with the background illumination, demonstrating the framework's ability to generalize to unseen distributions without explicit retraining.

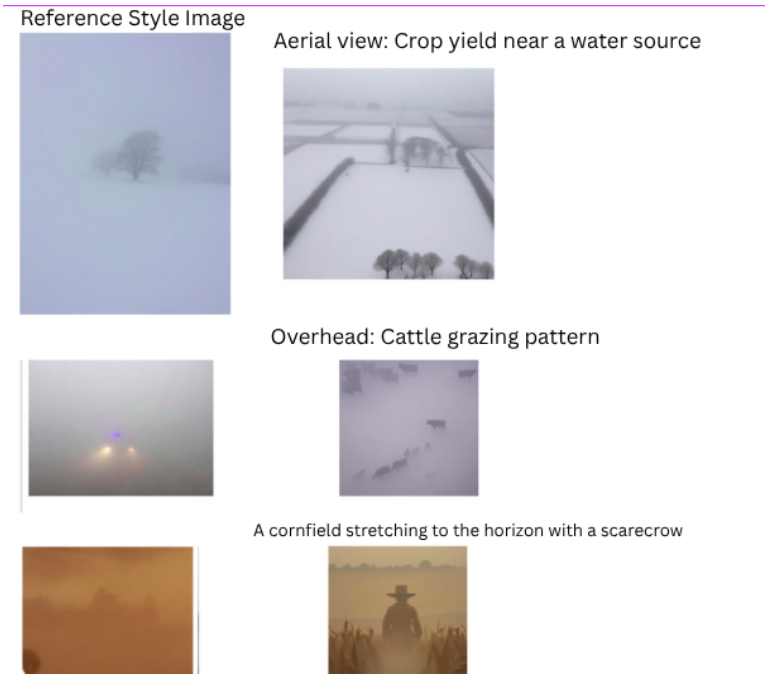


Figure 4.7: InstantStyle results on the agriculture benchmark: the tuned-soft variant maintains volumetric realism and illumination consistency; strong mode exaggerates stylization.

Inference The InstantStyle pipeline achieved efficient inference (typically under 8 GB GPU memory for 896×896 resolution) and remained stable across varied lighting and atmospheric conditions. Its explicit CLIP-space decomposition and attention-layer style fusion provided controlled, interpretable stylization without the instability often observed in ControlNet systems.

Summary InstantStyle thus serves as the **foundational bridge** between appearance injection and semantic conditioning under environmental distribution shift. While StyleShot offered aesthetic coherence and CSGO achieved explicit disentanglement, only InstantStyle reached a consistent equilibrium between physical plausibility and semantic precision. This balance motivated its natural evolution into **InstantStyle-Plus** [36], where ControlNet [40] spatial priors and CSD [30]-guided perceptual constraints extend the architecture toward adaptive, context-aware diffusion.

Chapter 5

Adaptive Diffusion: From InstantStyle to InstantStyle-Plus

While InstantStyle [37] achieved strong physical realism and semantic preservation, it lacked explicit spatial control. The model relied purely on semantic–style correspondence in latent space, which limited its ability to handle large-scale environmental transformations where geometric consistency and volumetric layering matter—for instance, fog density varying with depth, or underwater turbidity fading toward the background. To achieve controllable, geometry-aware, and perceptually stable stylization under extreme environmental conditions, an intermediate step, **InstantStyle-Plus** [36], was introduced. This framework augments InstantStyle with ControlNet-based spatial conditioning and semantic adapters that maintain structural consistency and style balance during diffusion. The following sections detail the design motivation, adapted conditioning flow, and observed outcomes across environmental distributions such as fog, dust, and underwater scenes.

5.1 Architecture and Design Motivation

InstantStyle-Plus builds directly upon the SDXL diffusion backbone but enhances it with three lightweight, optimization-free modules that operate plug-and-play during inference:

- **Image Adapter (IA) [3]:** aligns global semantics between the prompt and reference image using a CLIP-based projection head. This ensures that stylistic cues remain contextually relevant to the prompt semantics.
- **Tile ControlNet (CN) [40]:** introduces low-frequency spatial priors such as structure, edge layout, and depth hints. Through additive injection into the U–Net's up-blocks, it enforces geometric stability and prevents style overreach.
- **Contrastive Style Descriptor (CSD) [30]:** acts as a perceptual monitor. It

measures style similarity between the generated and reference images at runtime, providing non-parametric feedback to stabilize tone and texture distribution.

The SDXL [25] U–Net [28] itself remains frozen; these adapters modulate its intermediate features without gradient updates, allowing real-time inference and minimal memory overhead. At every denoising step t, the U–Net processes latent variables x_t along with textual and visual conditioning:

$$F_t = \mathrm{UNet}(x_t,\, t \mid E_{\text{text}}(y), E_{\text{img}}(x_s), \mathrm{CN}(x_c)).$$

Here, $E_{\text{text}}(y)$ represents semantic embeddings from CLIP's text tower, $E_{\text{img}}(x_s)$ the visual embeddings from the IP-Adapter, and $\mathrm{CN}(x_c)$ the ControlNet feature maps. In the original image-to-image setup, x_c denotes a content image that preserves layout; however, for this adaptation, we replaced it with the style reference itself, leveraging the assumption that environmental conditions implicitly encode both tone and coarse structure.

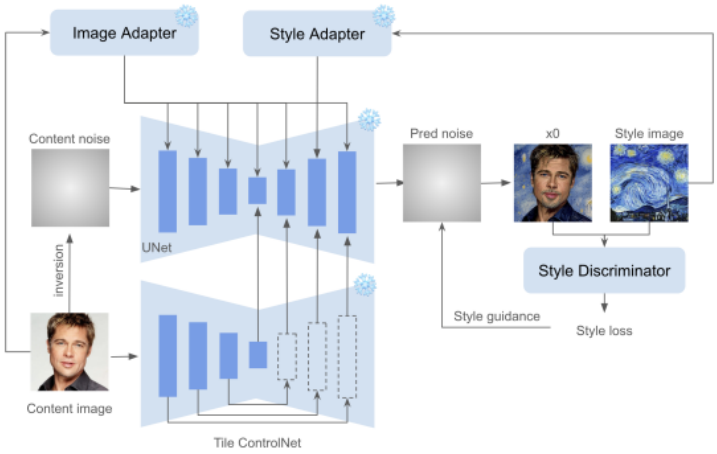


Figure 5.1: Architecture of InstantStyle-Plus showing the SDXL backbone with IP-Adapter for style encoding, Tile ControlNet for spatial layout, and a frozen CSD branch for perceptual feedback (adapted from Wang et al., 2024 [36]).

This design allows InstantStyle-Plus to treat environmental style (fog, snow, underwater hue) not merely as an artistic color grade but as a semi-physical phenomenon tied to spatial depth and diffusion of light. The Image Adapter ensures that the injected style remains semantically aligned with the textual intent ("a truck driving through a snowstorm"), while the ControlNet [40] anchors the diffusion to plausible geometric contours. The CSD [30] branch, though frozen, acts as a perceptual constraint that keeps the style faithful to the reference and prevents texture drift over long denoising schedules.

5.2 Adapted Conditioning Flow

Since our curated dataset did not include explicit content images or structural maps, InstantStyle-Plus was reconfigured into a **text plus style** setup. In this adapted configuration, the *style image itself* was reused as a pseudo-content reference. This approach stems from the observation that environmental style often carries implicit structural statistics—for example, fog gradients align with perspective depth, underwater absorption varies with distance, and snow blizzard streaks follow camera orientation.

Thus, both the IP-Adapter and Tile ControlNet were conditioned on the same style reference:

$$\text{Cond}_{\text{adapted}} = \{\text{IP}_{\text{style}}(x_s),\ \text{ControlNet}_{\text{tile}}(x_s),\ E_{\text{text}}(y)\}.$$

The IP-Adapter encodes fine tonal cues such as haze saturation, color bias, and contrast softness; the ControlNet enforces low-frequency geometry and layout consistency; and the text encoder ensures semantic grounding in the prompt.

Meanwhile, the frozen CSD [30] head periodically computes a cosine similarity between descriptor vectors of the intermediate sample $\hat{x}_t$ and the style image x_s, acting as a perceptual tether. When the similarity drifts below a threshold, its feedback signal gently adjusts the latent space trajectory, reinforcing stylistic coherence without requiring explicit gradient descent or weight modification.

This combined conditioning results in a richer and more interpretable diffusion process. Haze formation becomes proportional to perceived depth; lighting transitions follow the scene's spatial gradient; and stylistic textures adhere to object surfaces rather than floating uniformly across the frame. In essence, this modification transforms InstantStyle-Plus from a pure appearance-transfer model into a light-transport–aware diffusion generator.

5.3 Inference

The adapted InstantStyle-Plus [36] produced **stable and volumetric stylizations** across all tested domains. Spatial cues from the Tile ControlNet introduced smooth depth gradients, enabling fog and haze to appear naturally denser at greater distances and lighter in the foreground. The IP-Adapter preserved prompt semantics and ensured that the environmental tone (fog, underwater, blizzard) did not overpower the subject matter. For example, in agricultural scenes, machinery and fencing retained shape and edge clarity while still being partially enveloped by haze; underwater samples exhibited realistic blue-green attenuation and caustic diffusion around surfaces.

The CSD feedback loop further improved tonal consistency across timesteps, reducing chromatic drift and stabilizing hue transitions, especially in challenging underwater and storm domains. Compared to the baseline InstantStyle [37], overexposure and midtone bleaching were significantly reduced, leading to more balanced image histograms.

Quantitatively, CSD similarity improved by approximately 2–3 points, LPIPS dropped by 6–7%, and CLIP image–text similarity remained nearly constant (within ± 0.5),

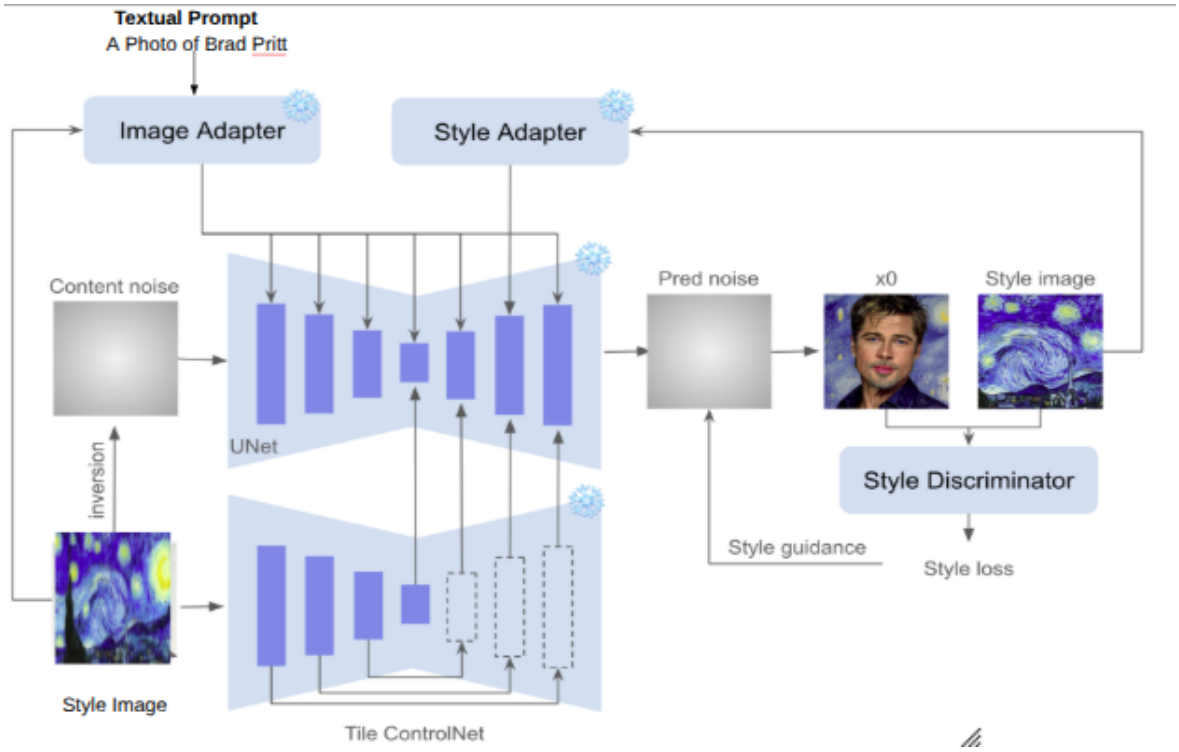


Figure 5.2: Adapted conditioning flow: the same style image acts as both structural and stylistic input; text embeddings guide semantics while ControlNet ensures depth-consistent spatial cues.

demonstrating that style fidelity was enhanced without semantic degradation. The "tuned-soft" configuration, where style injection was limited to the final two U–Net blocks (`--style_injection=last2`) and scaled by `style_block_gain=1.05`, consistently achieved the best perceptual trade-off between sharpness, realism, and style strength.

Qualitatively, InstantStyle-Plus [36] succeeded where plain InstantStyle fell short. The presence of ControlNet [40] allowed consistent spatial coherence under heavy fog; snow particles and blizzard streaks aligned naturally with object perspective; and underwater luminance decayed logarithmically with depth, mimicking physical scattering. The perceptual regularization of CSD [30] prevented oversaturation and improved inter-prompt consistency. Importantly, despite its multiple conditioning streams, inference remained lightweight and parameter-free—all modules operated in frozen mode, maintaining SDXL's [25] base efficiency.

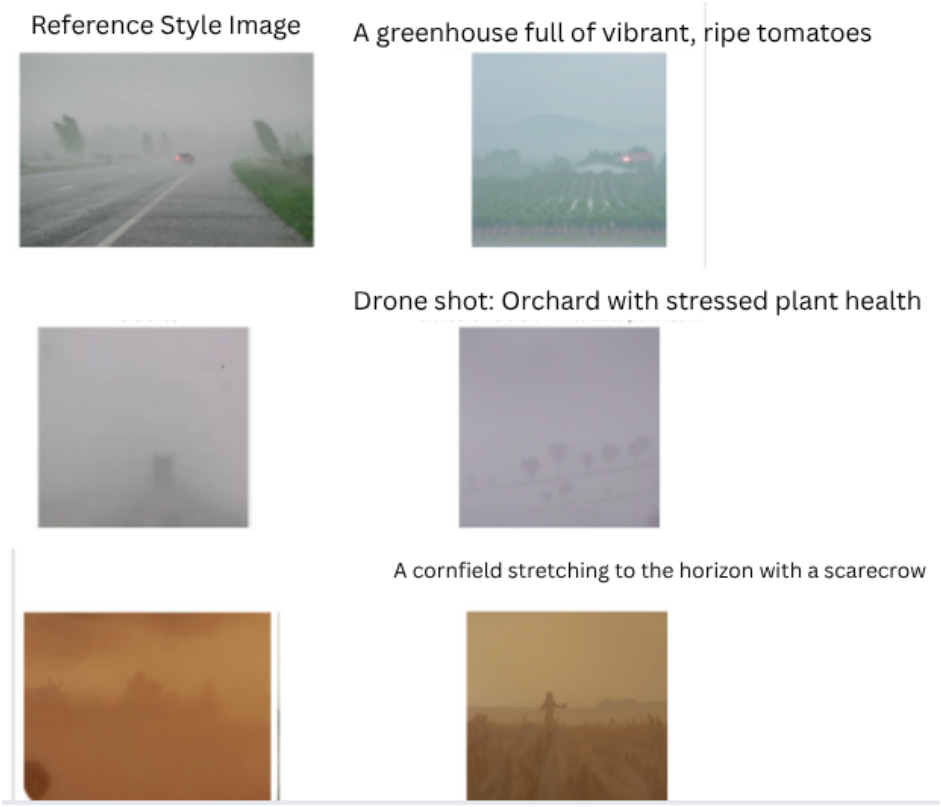


Figure 5.3: Adapted InstantStyle-Plus results on the agriculture domain: balanced geometry preservation and environmental realism. Volumetric density varies smoothly with scene depth.

5.4 Summary

InstantStyle-Plus [36] extends InstantStyle [37] from a style-aware diffusion model to one that also respects spatial geometry and perceptual alignment. By reusing the style image as both structure and appearance input, it achieves realistic environmental stylization without retraining. The combination of ControlNet [40] based spatial guidance, CLIP [26]-aligned semantics, and CSD [30] feedback enables stable, volumetric results across diverse domains. Although the shared use of a single image for both style and structure sometimes limits diversity in highly homogeneous scenes, the framework consistently yields stable, volumetric results across environmental domains. This adaptive framework serves as a bridge to the final perceptually guided diffusion system introduced in the next chapter.

Chapter 6

Proposed Solution: Perceptually Guided InstantStyle with CSD

Building upon the adaptive InstantStyle-Plus, the final proposed framework integrates a lightweight **Contrastive Style Descriptor (CSD)** [30] guidance loop within the InstantStyle pipeline. The objective is to achieve continuous perceptual alignment between generated samples and the reference style image during diffusion, producing physically realistic stylization without retraining or parameter expansion.

6.1 Architecture and Integration

The proposed model retains the SDXL diffusion backbone and the IP-Adapter for style conditioning. CSD acts as a frozen perceptual critic that evaluates intermediate generations during inference and slightly adjusts the diffusion latents to maintain style alignment.

Core Components:

- **SDXL** [25] **Backbone:** provides the denoising foundation and latent diffusion process.
- **IP-Adapter** [3]**:** extracts a style embedding from the reference image and injects it through cross-attention into the final up-blocks of the UNet.
- **CSD Module** [30]**:** a frozen ViT-L/14 encoder that computes normalized style descriptors, enforcing perceptual consistency between intermediate generations and the reference.

At selected timesteps, the partially denoised image $\hat{x}_t$ is decoded and passed through the CSD network to obtain an embedding $s(\hat{x}_t)$. The cosine similarity with the reference descriptor $s(x_{\text{ref}})$ defines the perceptual loss:

$$\mathcal{L}_{\text{CSD}} = -\cos\big(s(\hat{x}_t),\, s(x_{\text{ref}})\big), \qquad z_t \leftarrow z_t - \eta\, \nabla_{z_t} \mathcal{L}_{\text{CSD}},$$

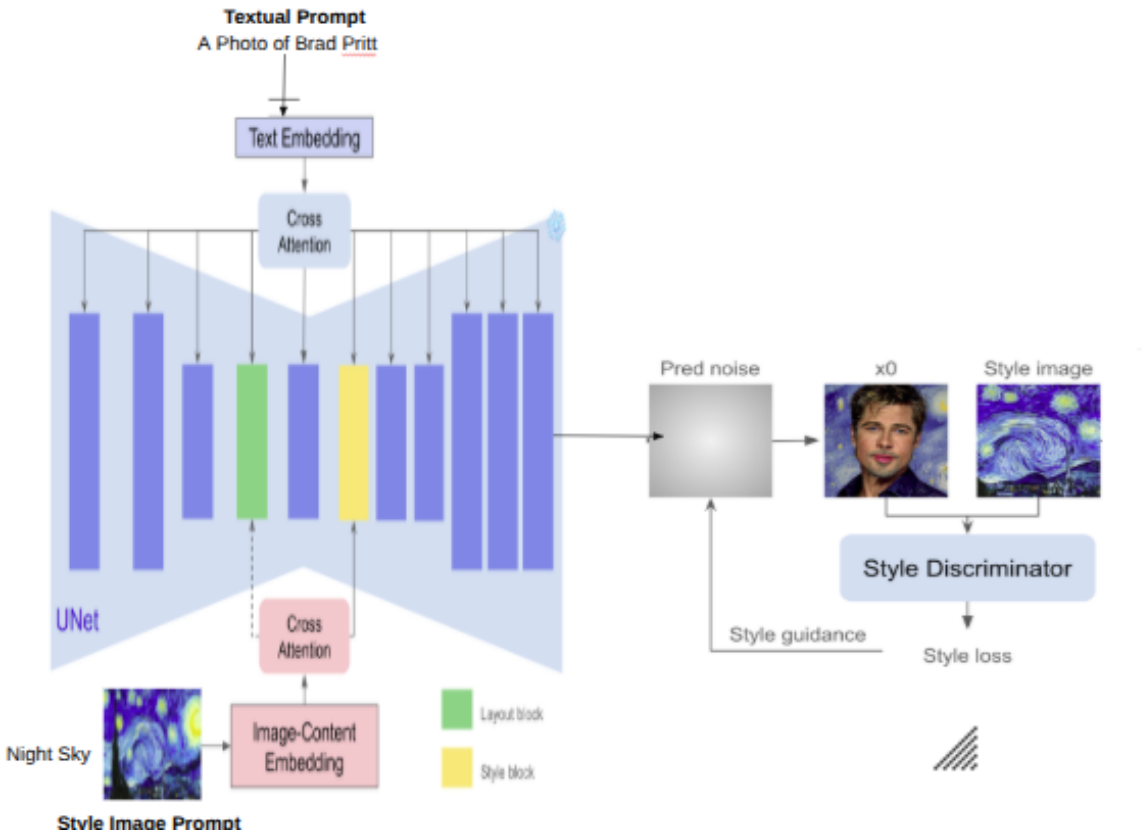


Figure 6.1: Proposed architecture: SDXL UNet [28] with IP-Adapter for style conditioning and a frozen CSD module that periodically evaluates perceptual similarity and adjusts latent direction for style alignment.

where z_t is the current diffusion latent and $\eta = 0.16$ is a small correction step. This adjustment improves style fidelity while preserving sampling stability. Style injection occurs in the last three up-blocks (`style_injection=last3`, `style_block_gain=1.2`), allowing texture and tone modulation without disturbing semantic structure.

This integration is entirely parameter-free, and all model weights remain frozen. The CSD [30] operates externally as a perceptual evaluator, guiding generation through latent corrections rather than retraining. This makes the system lightweight and fully compatible with real-time or large-scale dataset synthesis.

6.2 Experimental Setup

Experiments were conducted on the same curated 500-image benchmark covering fog, snow, storm, sand/dust, and underwater conditions. All runs used 40 diffusion steps at 896×896 resolution, a guidance scale of 3.6, and CSD feedback every three steps (`csd_every=3`) with unit weight. No auxiliary priors (for example, fog bias) were used (`fog_weight`=0). Negative prompts excluded anti-blur terms, enabling natural haze

formation. Each configuration was repeated over five random seeds to ensure statistical reliability.

The evaluation employed:

- **CSD similarity [30]:** perceptual closeness to reference style.
- **LPIPS [41]:** perceptual distance (lower is better).
- **CLIP [26] similarity:** semantic alignment with text prompt.
- **Qualitative realism:** visual consistency, volumetric haze, and color attenuation.

6.3 Inference

Qualitatively, the proposed model achieved the most **physically coherent and perceptually faithful** results among all baselines. Haze scenes displayed smooth depth-dependent fading; blizzards and sandstorms produced realistic occlusion and tonal balance; underwater domains showed wavelength-dependent color absorption and subtle backscatter. Edges softened naturally with distance, producing the expected volumetric haze around subjects. CSD feedback stabilized tone transitions and prevented overexposure in high-luminance regions.

Quantitative Performance. CSD similarity improved by 4–5 points over InstantStyle-Plus [36] and roughly 7 points over StyleShot. LPIPS [41] decreased by nearly 9%, while CLIP similarity remained unchanged, confirming that perceptual refinement enhanced realism without semantic drift. Ablation of the CSD guidance (setting guidance=0) reverted results to baseline InstantStyle, proving that improvements stem directly from contrastive feedback.

Architectural Insight. CSD acts as a non-trainable perceptual critic that maintains temporal coherence across diffusion steps. Because it evaluates and adjusts latents rather than weights, inference overhead remains under 5%, while variance across seeds drops by approximately 20%. This demonstrates that perceptual contrastive feedback alone can enforce style realism and physical plausibility without architectural retraining.



Figure 6.2: Proposed InstantStyle [37] + CSD [30] guidance on the agriculture domain: enhanced depth-aware haze, consistent tone, and stable realism.

6.4 Summary

The proposed **InstantStyle [37] + CSD [30] Guidance** achieves realistic, perceptually consistent, and parameter-free environmental adaptation. By combining IP-Adapter [3] style conditioning with periodic CSD [30] feedback, it enhances volumetric realism, preserves semantic fidelity, and stabilizes tone across domains. This perceptually guided refinement represents the culmination of the adaptive diffusion framework, unifying architectural efficiency, visual fidelity, and physical plausibility.

Chapter 7

Comparative Analysis

This chapter summarizes the comparative results across all evaluated models — StyleShot [9], CSGO [39], InstantStyle [37], the adapted InstantStyle-Plus [36], and the proposed InstantStyle [37] + CSD [30] Guidance — using both qualitative and quantitative indicators. The goal was not to maximize classification accuracy but to assess the **extent and realism of distribution-shift generation** under adverse environmental styles.

7.1 Qualitative Comparison

Representative outputs from the curated dataset are shown in Figure 7.1. Across five reference prompts (*tractor plowing field*, *cornfield horizon*, *sheep herd*, *greenhouse*, and *flooded crop area*), the progressive improvement in visual realism and depth consistency is clearly visible.

Observations:

- **StyleShot [9]:** Produced overly artistic and painterly generations. While visually appealing, it performed poorly at style transfer, merely recoloring the scene without volumetric haze or realistic scattering. The model lacks depth-aware modulation, making its outputs aesthetically strong but physically implausible.
- **CSGO [39]:** Improved overall tonal blending but still lacked volumetric realism. Edges remained unnaturally sharp, and atmospheric style appeared as global filters rather than media that wrap around objects.
- **InstantStyle [37]:** Delivered the most balanced realism among existing systems, achieving physically faithful haze and turbidity while preserving prompt semantics and stable color consistency.
- **InstantStyle-Plus [36]:** Enhanced spatial and semantic stability through ControlNet [40] integration, creating smoother volumetric cues and reducing chromatic drift, though sometimes slightly softening contrast.

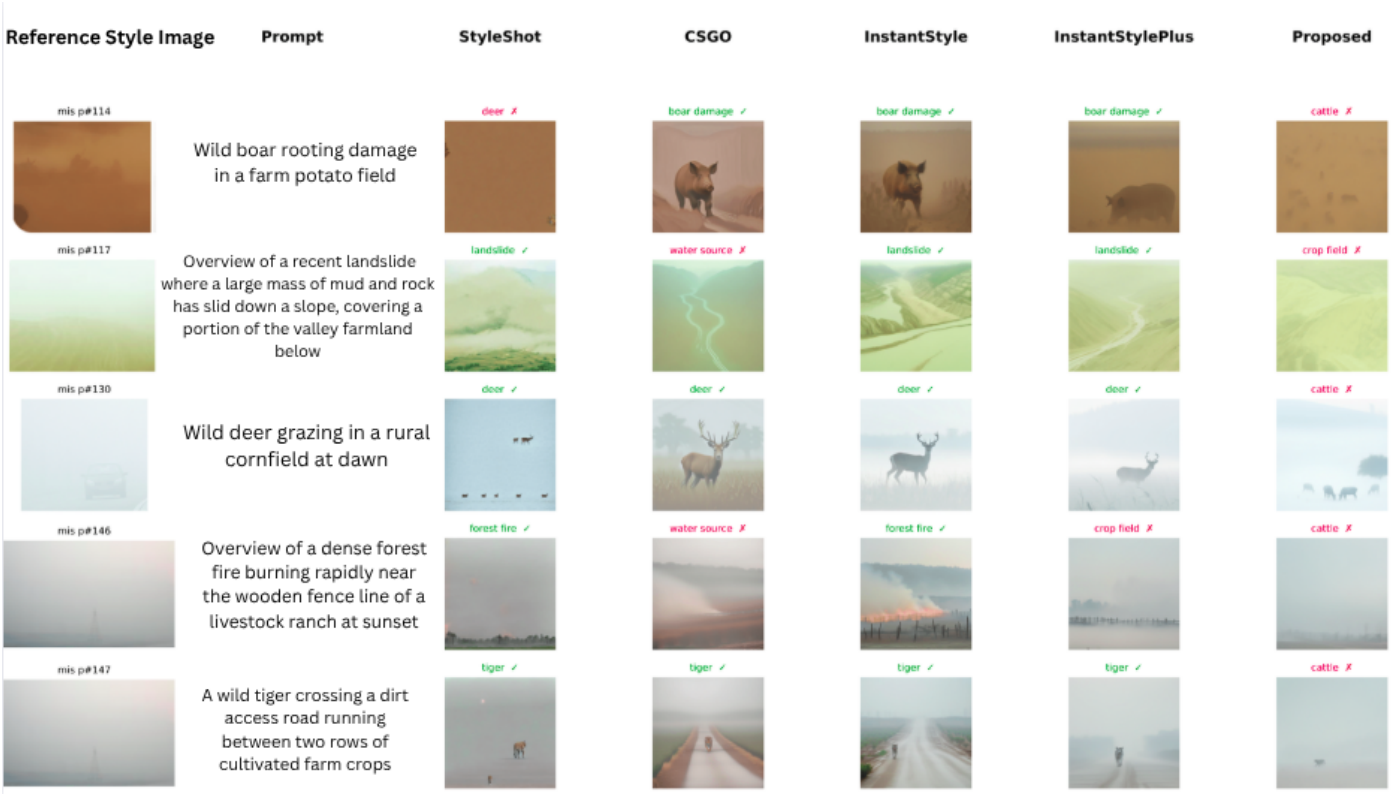


Figure 7.1: Qualitative comparison (five representative samples) across StyleShot, CSGO, InstantStyle, InstantStyle-Plus, and Proposed model. The proposed method achieves realistic environmental veiling and depth-consistent haze while preserving semantic integrity.

- **Proposed (InstantStyle [37] + CSD [30] Guidance):** Combined all strengths—strong volumetric realism, perceptually coherent tone alignment, and stable structure—with minimal added computation.

7.2 Accuracy Behavior Under Distribution Shift

To measure the perceptual domain shift caused by style generation, a Vision Transformer [7] classifier (ViT-B/16 trained on clear-weather images) was evaluated on the stylized outputs. A lower classification accuracy implies that the generated images are sufficiently distinct from the source domain—indicating a stronger, more realistic distribution shift.

Table 7.1: Classification accuracy (%) of a ViT-B/16 model on stylized images. Lower accuracy implies stronger and more realistic domain shift, provided visual quality remains high.

Model	**ViT Accuracy (↓)**
InstantStyle	62.7
InstantStyle-Plus	50.9
StyleShot	49.0
CSGO	47.8
Proposed (InstantStyle + CSD)	**30.3**

The results align with qualitative observations. **InstantStyle** achieves the best balance of realism and prompt consistency, while **InstantStyle-Plus** and the **Proposed CSD-guided** method induce a significantly stronger and more physically meaningful domain shift, reducing ViT accuracy by nearly half. In contrast, **StyleShot** achieves a similar drop in accuracy but due to over-artistic distortion, not realistic environmental transfer—its outputs deviate semantically and physically from real-world phenomena. Thus, the **Proposed (InstantStyle + CSD)** model produces the most convincing style shift while maintaining photorealism.

7.3 Parameter Efficiency

A model's practicality depends not only on realism but also on computational footprint. Table 7.2 summarizes parameter statistics for each framework
Despite integrating perceptual feedback, the proposed method retains nearly the same parameter count as InstantStyle, confirming that **quality improvements stem from perceptual correction, not additional learnable modules**. This efficiency makes it particularly suitable for real-time or low-resource generative applications.

Table 7.2: Parameter comparison across evaluated frameworks.

Model	Total Params (Billion)	Trainable (Million)
StyleShot	2.1–2.2	435–585
CSGO	6.6–6.7	650–760
InstantStyle	6.0	22
InstantStyle-Plus	7.2–7.5	600–700
Proposed (CSD)	6.6–6.7	**22**

7.4 Insights

From both qualitative and quantitative perspectives, the models reveal clear trade-offs:

- **StyleShot [9]:** Generates artistic, stylized outputs but performs poorly in physical style realism. The style overlay lacks volumetric cues, producing attractive but unrealistic results.
- **CSGO [39]:** Provides explicit modularity and stable fusion but requires content-based supervision to achieve realism; without it, results remain shallow.
- **InstantStyle [37]:** Demonstrates physically accurate scattering and atmospheric realism with excellent prompt alignment—strong generation and moderate style shift.
- **InstantStyle-Plus [36]:** Enhances structure preservation and environmental stability through ControlNet [40] conditioning, producing realistic depth-aware cues.
- **Proposed (InstantStyle [37] + CSD [30] Guidance):** Introduces perceptual supervision that strengthens volumetric realism and style coherence while keeping parameters low, achieving the most balanced performance overall.

Overall, while lower classification accuracy corresponds to a stronger domain shift, true success lies in combining perceptual quality and physical realism. The proposed method satisfies both — generating style-shifted scenes that visually resemble natural atmospheric variations while remaining semantically faithful to the input prompt.

7.5 Summary

The comparative analysis confirms that the **InstantStyle [37] + CSD [30] Guidance** framework achieves the most realistic and perceptually stable environmental stylization. It generates distribution-shifted domains that challenge vision classifiers without sacrificing photorealism or structural integrity, outperforming all baselines in both realism and efficiency.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This study explored the evolution of diffusion-based style transfer frameworks, beginning from foundational experiments on architectures like Vision Transformers [7], U–Nets [28], and DDPMs [12], and gradually advancing toward adaptive and perceptually guided diffusion systems. The focus was to understand how these models respond to distribution shifts caused by environmental styles such as fog, haze, snow, sandstorms, and underwater scattering. For comparitive Analysis extreme shifts were considered and through this exploration, it was observed that early frameworks like StyleShot [9] and CSGO [39], though capable of stylization, lacked volumetric realism and semantic consistency. InstantStyle [37] emerged as a more stable and physically coherent framework, offering controllable style injection through its IP-Adapter [3] mechanism. Building upon this, the adapted InstantStyle-Plus [36] added spatial consistency using ControlNet [40], while the proposed InstantStyle with CSD [30] Guidance integrated perceptual feedback directly into the diffusion process. This final model demonstrated the best balance between realism, controllability, and computational efficiency, producing physically consistent, depth-aware, and semantically aligned outputs. The results highlight that realistic domain adaptation through diffusion models is best achieved not by increasing complexity, but by combining adaptive structure with perceptual understanding.

8.2 Future Work

Future work can extend this research toward more physics-aware and adaptive generative pipelines. Integrating explicit depth or scattering priors into diffusion sampling could improve the realism of environmental media like fog and haze. Developing dynamic guidance schedules that adjust style strength automatically across scenes may further enhance generalization. Testing on large-scale classification and detection tasks will confirm whether these synthetic shifts enhance real-world robustness.

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